PROJECT LOGIC

Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Platform	ServiceNow
Core Objective	Enforce record-level access based on user roles and the Branch field value.
Scope	READ, CREATE, WRITE and DELETE ACLs for the Institution Details table.

1. Project Objective

The project demonstrates how ServiceNow script-controlled Access Control Lists (ACLs) can restrict access to records according to user roles and record values. The source scenario allows EEE users to access EEE branch records while administrators retain full access. Additional roles are then used to control create, write and delete operations.

2. Project Logic / Functional Flow

1. 1. Create the EEE test user and the custom roles bb1, bb2, bb3 and bb4.
2. 2. Create the custom Institution Details table (u_institution_details).
3. 3. Add student, faculty, branch, email, phone and description fields.
4. 4. Create sample records containing ECE, EEE and CSE branch values.
5. 5. Configure a READ ACL for the table with the Branch = EEE condition and an advanced script.
6. 6. The READ script grants access to administrators and users with the bb1 role; other users are denied.
7. 7. Configure CREATE with bb2, WRITE with bb3 and DELETE with bb4.
8. 8. Test the combined permissions through user impersonation and the table list.
9. 9. Verify that the final permission set permits the intended operations only for the allowed EEE records.

3. Data and Security Objects

Object	Configuration	Purpose	Expected Result
User	EEE User	Test role-based access	User receives assigned custom roles
Roles	bb1, bb2, bb3, bb4	Separate operation permissions	Permissions can be combined
Table	u_institution_details	Store institution/student records	Records are protected by ACLs
Branch	ECE / EEE / CSE	Record-level access condition	READ access is restricted to EEE in the source logic
ACL	READ	Control viewing	bb1 users can read EEE records; admin has full access
ACL	CREATE	Control record creation	bb2 is required
ACL	WRITE	Control record modification	bb3 is required
ACL	DELETE	Control record deletion	bb4 is required

4. READ ACL Decision Logic

Source script logic:

```
(function () {
  if (gs.hasRole('admin')) {
    return true;
```

```
  }

  if (gs.hasRole('bb1')) {
    return true;
  }

  return false;
})();
```

The source document also specifies a data condition of Branch is EEE. Therefore, the intended READ behavior is: administrator → full access; bb1 user → EEE records; user without the required role → no records.

5. Permission Matrix

User/Role	READ	CREATE	WRITE	DELETE	Source Verification
Administrator	All records	All records	All records	All records	Explicitly stated for READ; final outcome describes admin protection context
bb1	EEE records	No	No	No	READ ACL
bb1 + bb2	EEE records	Yes	No	No	CREATE ACL verification
bb1 + bb2 + bb3	EEE records	Yes	Yes	No	WRITE ACL verification
bb1 + bb2 + bb3 + bb4	EEE records	Yes	Yes	Yes	DELETE ACL verification
No required role	No records	No	No	No	READ verification and role-based ACL design

6. Verification Logic

- Impersonate the EEE user and open the Institution Details list.
- Confirm that only EEE branch records are visible under the READ restriction.
- Impersonate a user without the required role and confirm that records are not displayed.
- Impersonate an administrator and confirm visibility of records regardless of branch.
- Add bb2 and verify the New button/create capability.
- Add bb3 and verify that EEE records can be edited.
- Add bb4 and verify that EEE records can be deleted.
- Use the table name u_institution_details.list/LIST to open the list during testing.

7. End-to-End Project Logic

User Identity → Assigned Roles → ACL Operation → Branch/Data Condition → Script Evaluation → Allow/Deny Decision → UI/List/Form Behavior.

- Identity establishes who is requesting access.
- Roles determine whether the user is eligible for an operation.
- The ACL identifies the operation: read, create, write or delete.
- The Branch condition restricts record visibility to EEE for the READ scenario.
- The advanced script evaluates the administrator and bb1 conditions.
- ServiceNow returns an allow or deny decision and the interface reflects the resulting permission.

8. Project Outcome

The source project states that learners gain understanding of script-controlled READ, WRITE, CREATE and DELETE ACLs, how user roles and record field values are evaluated together, and how ACLs protect sensitive data from unauthorized viewing, modification, creation and deletion.